

# INTERIOR AND BOUNDARY POINTS OF $c_{00}$ AND ITS COMPLEMENT

**Definitions.**

$$\ell^\infty = \{x = (x_n)_{n=1}^\infty : \sup_{n \in \mathbb{N}} |x_n| < \infty\}, \quad d_\infty(x, y) = \sup_{n \in \mathbb{N}} |x_n - y_n|,$$

$$c_{00} = \{x \in \ell^\infty : x_n \neq 0 \text{ for only finitely many } n\},$$

$$c_0 = \{x \in \ell^\infty : x_n \rightarrow 0 \text{ as } n \rightarrow \infty\}.$$

We write  $A^c$  for the complement of a set  $A$ ,  $\text{Int}(A)$  for its interior,  $\text{Bd}(A)$  for its boundary and  $\text{Cl}(A) = \text{Int}(A) \cup \text{Bd}(A)$  for its closure.

**1.**  $\text{Int}(c_{00}^c) = c_0^c$ .

*Proof.* (a) Show  $c_0^c \subseteq \text{Int}(c_{00}^c)$ .

Let  $x \in c_0^c$ . Then  $x \notin c_0$ , so there exists  $\delta > 0$  such that

$$\#\{n : |x_n| \geq \delta\} = \infty.$$

Set  $\varepsilon = \delta/2$ . If  $y \in B(x, \varepsilon)$  then

$$|y_n| \geq |x_n| - d_\infty(x, y) > \delta - \varepsilon = \delta/2 > 0$$

for all infinitely many  $n$ , so  $y \notin c_{00}$ . Hence  $B(x, \varepsilon) \subset c_{00}^c$ , and  $x \in \text{Int}(c_{00}^c)$ .

(b) Show  $\text{Int}(c_{00}^c) \subseteq c_0^c$ .

Let  $x \in c_0$ . For any  $\varepsilon > 0$ , choose  $N$  so that  $|x_n| < \varepsilon$  for all  $n \geq N$ . Define

$$y_n = \begin{cases} x_n, & n < N, \\ 0, & n \geq N. \end{cases}$$

Then  $d_\infty(y, x) < \varepsilon$  but  $y \in c_{00}$ , so no ball around  $x$  lies in  $c_{00}^c$ . Thus  $x \notin \text{Int}(c_{00}^c)$ .

Combining (a) and (b) yields

$$\text{Int}(c_{00}^c) = c_0^c.$$

□

**2.**  $\text{Int}(c_{00}) = \emptyset$  and  $\text{Bd}(c_{00}) = c_0$ .

*Proof.* (a) Show  $\text{Int}(c_{00}) = \emptyset$ .

Let  $x \in c_{00}$  with support in  $\{1, \dots, N\}$ . Given  $\varepsilon > 0$ , define

$$y_n = \begin{cases} x_n, & n \leq N, \\ \frac{\varepsilon}{2}, & n > N. \end{cases}$$

Then  $d_\infty(y, x) < \varepsilon$  but  $y \notin c_{00}$ . Hence no open ball around  $x$  lies in  $c_{00}$ , so  $\text{Int}(c_{00}) = \emptyset$ .

(b) *Show*  $\text{Bd}(c_{00}) = c_0$

Recall the general relation

$$\ell^\infty = \text{Int}(c_{00}) \cup \text{Bd}(c_{00}) \cup \text{Int}(c_{00}^c).$$

We have already established that  $\text{Int}(c_{00}) = \emptyset$  and  $\text{Int}(c_{00}^c) = c_0^c$ . Hence, we conclude that  $\text{Bd}(c_{00}) = c_0$ .

□